

Data Driven Soccer Recruitment System

Identifying Undervalued Talent Through Machine Learning

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Abstract

This project aims to develop an AI-powered soccer recruitment system that helps clubs identify undervalued players and potential transfer targets through machine learning and data analytics. The system will leverage player performance data, soccer metrics, and market valuation information to predict player market values, identify players statistically similar to one another, and uncover recruitment opportunities that traditional scouting methods may overlook. Multiple machine learning models, including Linear Regression, Random Forest, and XGBoost, will be trained and evaluated. The project will also incorporate clustering techniques, feature-importance analysis, and Tableau dashboards to provide actionable insights for recruitment decisions. The project will create a soccer recruitment tool that helps teams find talented players who may be worth more than their current market price. Using ML, the system will predict player value and identify undervalued players to support recruitment decisions.

Project Description

Professional soccer clubs invest billions of dollars annually in player recruitment, scouting, and transfer activities. Identifying talented players before their market value significantly increases can provide a competitive advantage and generate substantial financial returns. But evaluating players is a complex task that requires analyzing hundreds of performance metrics, understanding contextual factors, and forecasting future potential. Traditional scouting methods rely heavily on subjective observations and expert judgment, which can introduce bias and inconsistency into recruitment decisions.

Advancements in data science and machine learning have transformed the way professional sports organizations evaluate talent. Modern clubs increasingly rely on statistical models, predictive analytics, and advanced performance metrics to supplement traditional scouting

practices. Metrics such as Expected Goals (xG), Expected Assists (xA), progressive passes, successful dribbles, defensive actions, and possession-based statistics provide a more comprehensive understanding of player performance than traditional statistics alone.

The primary objective of this project is to develop a machine learning system capable of predicting player market values based on performance statistics and identifying players whose predicted value exceeds their current market value. These players may represent cost-effective transfer opportunities for clubs seeking to maximize return on investment.

The project focuses on three main tasks:

- Player Market Value Prediction: Use a player's performance, age, position, league, and playing time to predict their market value using machine learning models
- Player Similarity Analysis: using clustering and dimensionality. Group players with similar playing styles and statistics so clubs can find alternative transfer targets at a lower cost.
- Undervalued Talent Identification: Compare the player's current market value with the predicted market value to identify and rank undervalued players who could be good transfer targets.

Soccer clubs such as Liverpool, Brighton, and Brentford have demonstrated how data-driven recruitment strategies can improve player acquisition efficiency and competitive performance. Also, the 2026 FIFA World Cup provides a timely opportunity to evaluate emerging talent and apply the developed models to real-world player scouting scenarios. The project will utilize two primary publicly available datasets, FBref and Transfermarkt, and one data acquisition tool.

1. The Transfermarkt dataset supplies the prediction target and contextual variables: 37,000+ player profiles with current market values, 500,000+ historical valuation records, 99,000+ transfers with fees, per-game appearances, and national-team data, including FIFA rankings.

2. An FBref-sourced top-five-league statistics dataset supplies the model's performance features: advanced per-90 metrics such as Expected Goals (xG), Expected Assists (xA), progressive passes and carries, key passes, tackles, interceptions, touches, and goalkeeping statistics.

3. Soccerdata, an open-source Python library, serves as a programmatic acquisition layer that can pull and refresh FBref, Transfermarkt, and Understat data into consistent tables and help align player identifiers across sources.

The dataset will include player data from the top five European leagues over multiple seasons. It is expected to contain approximately 15,000-20,000 player records, each with over 100 performance features. This provides enough data to train and evaluate machine learning models, identify similar players, and detect undervalued talent. The cleaned data will be stored in a PostgreSQL database. Machine learning models, including Linear Regression, Random Forest, and XGBoost, will be used to predict player market values. The predicted values will then be compared with actual market values to identify undervalued players. K-Means clustering and Principal Component Analysis (PCA) will also be used to group players with similar performance profiles and recommend alternative transfer targets.

System Pipeline: Data Sources → Data Collection and Integration → Data Cleaning and Feature Engineering → PostgreSQL Database → Exploratory Data Analysis → Machine Learning → Player Similarity Analysis → Model Explainability → Undervalued Talent Identification → Tableau Dashboards → Soccer Recruitment Recommendation System

References

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Datasets:

<https://www.kaggle.com/datasets/hubertsidorowicz/football-players-stats-2025-2026.csv>

<https://www.kaggle.com/datasets/davidcariboo/player-scores/data>

<https://pypi.org/project/soccerdata/>